Anup Sedhain

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Experience

Sr. Fullstack AI Engineer, Airvet Inc. – Los Angeles, CA Jan 2026 – Present

- Own and maintain Airvet's vet-tech Care Portal built with Next.js, TypeScript, and Go, partnering with the operations team to triage production issues and ship reliability improvements including redesigned SSE architecture with exponential backoff, and Firebase Cloud Messaging notifications.
- Build veterinary case classification systems using Python and Claude API with chain-of-thought prompting and contrastive few-shot examples; exploring PetBERT fine-tuning and RAG pipelines to further reduce inference cost and improve diagnostic accuracy.

Software Engineer, Airvet Inc. – Los Angeles, CA May 2025 – Dec 2025

- Led development of Airvet's parent-facing web application using NextJs, Golang, Twilio, and AWS and spearheaded research into veterinary AI diagnostic tools leveraging RAG architecture with Langchain.
- Architected AI-driven automation using Model Context Protocol (MCP) to streamline end-to-end regression testing workflows, reducing QA time by 80% and accelerating feature-to-release cycles from 4 weeks to 2 weeks.
- Engineered end-to-end automated development pipeline integrating JIRA, Figma, and GitHub with MCP agents, enabling automated releases while maintaining 99.9% deployment success rate.

Graduate Research Assistant, University of Florida – Gainesville, FL Jan 2025 – Jun 2025

- Developed a multistage social analysis pipeline processing 2M+ BlueSky events on the HiPerGator HPC cluster, combining BERT, K-means and TF-IDF for 92% accurate AI content detection with temporal network analysis (NetworkX, PageRank) to model attitude evolution.

Software Engineer, Airvet Inc. – Los Angeles, CA May 2021 – Jul 2024

- Engineered a real-time Care Portal with React, Redux, and Golang, and optimized system performance using Server-Sent Events and Redis pub-sub; enabled vet-techs to handle 100+ concurrent cases, accelerating response times by 75% while maintaining sub-200ms latency and 99.9% uptime.
- Spearheaded the modernization of airvet.com, collaborating with product, marketing, and design to replace a legacy system with a Next.js application, leveraging WordPress as a headless CMS and implementing ISG.
- Owned the feature roadmap for key B2B products by building an analytics ecosystem (Firebase, Amplitude, LogRocket) to inform development; led the creation of new admin tools that empowered teams with data insights, directly contributing to product iterations that resulted in almost 100% client retention.

Software Engineer, Freelance, Polydelic AS – Oslo, Norway Aug 2020 – Jun 2021

- Delivered 6+ SaaS platforms using React, Node.js, GraphQL, and AWS through cross-functional collaboration with product and design team, helping secure millions in funding and serving 500K+ users with 99.9% uptime.
- Engineered a video streaming platform by implementing a mediasoup-based Selective Forwarding Unit (SFU), achieving scalability to support 100+ concurrent users through strategic load balancing and Redis caching.
- Launched artist booking marketplace (Rubato) utilizing ElasticSearch and automated contract processing with GraphQL APIs, facilitating €100K+ monthly transactions.

Software Engineer, Leapfrog Technology Inc. – Kathmandu, Nepal Jul 2019 – Dec 2020

- Spearheaded enterprise project management platform development using Firebase and LitElement with automated workflows, accelerating delivery timelines by 40% for 50+ projects.
- Helped build bio-informatics application managing protein workflows with MySQL database and Micronaut, supporting 200+ batches monthly and improving processing efficiency by 25%.

Education

University of Florida, Master of Science in Computer Science, GPA 4.0 — Aug 2024 – Dec 2025

Kathmandu University, Bachelor of Computer Engineering, GPA 3.63 — Sep 2015 – Jul 2019

Technical Skills

AI/ML: MCPs, RAG, Langchain, Vector Databases, Hugging Face, Ollama, TGNNs, ML Pipelines
Programming: Python, TypeScript, JavaScript, Go, C++, Node.js, Java, Rust, OpenMP, MPI
Frontend: React, Next.js, Redux, Tailwind CSS, WebRTC, WASM, Real-time Systems
Backend & Data: PostgreSQL, MongoDB, Redis, Firebase, ElasticSearch, Spring Boot, RESTful APIs, GraphQL
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Kafka